

Batch Normalization

scaling the logits it will provide following advantages

- *Vanish gradient problem will reduce*
- *Convergence (reach to minimum point) fast*
- *Avoid overfitting situations*
- *mean = 0 std = 1*

$$w_1 * x + b = z$$

$$\mu = \frac{\sum_{i=1}^n z_i}{n} \quad , \quad \sigma = \sqrt{\sum_{i=1}^n (z_i - \mu)^2}$$

$$z' = \frac{z - \mu}{\sigma}$$

$$Z_{norm} = \gamma z' + \beta$$

Inputs === logits === Batch Norm == activation